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RBM20 Truncating Variants and Human Cardiomyopathy

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IMPORTANCE Genetic diagnosis has become increasingly important to guide clinical decision-making for patients with dilated cardiomyopathy (DCM). Pathogenic or likely pathogenic (P/LP) missense variants in the gene *RBM20* cause a highly penetrant arrhythmogenic DCM, but the role of *RBM20* truncating variants (*RBM20*tvs) is unclear.

OBJECTIVE To assess the contribution of *RBM20* variants to arrhythmogenic DCM.

DESIGN, SETTING, AND PARTICIPANTS In this cohort study, participants in the genome-first UK Biobank (UKB) and All of Us populations were evaluated to assess the etiologic fraction, natural history and penetrance of *RBM20* variants. Retrospective data were collected from an international cohort of patients with DCM and *RBM20* variants identified at centers of excellence for genetic heart disease and compared based on time to event. Study dates are not disclosed because the institutional review board did not authorize the sharing of this information.

EXPOSURES *RBM20* variants were compared to known P/LP variants and variants of uncertain significance in *RBM20* as well as titin truncating variants (*TTN*tvs).

MAIN OUTCOMES AND MEASURES Major ventricular arrhythmias, end-stage heart failure, and heart failure hospitalization as measured by medical record review (retrospective cohort) and diagnostic codes (UKB).

RESULTS Two main cohorts were studied for this project. In UK Biobank, a cohort of participants with *RBM20*tvs, *RBM20* synonymous variants, and *TTN*tvs was studied. Of these 4249 participants, 1869 (44%) were male. The mean (SD) age at enrollment was 56 (8.2) years. In the RBM20 registry, of 179 patients, 105 (58.6%) were male, and the mean (SD) age at enrollment was 43.8 (19.1) years. A validation cohort from the All of Us biobank was also used. This consisted of 7002 participants, 4342 of whom (62.0%) were male, and the mean (SD) age was 52.7 (16.7) years. The etiologic fraction of *RBM20* variants in arrhythmogenic DCM was 0.53 (95% CI, 0.32-0.67; $P < .001$). In genome-first biobanks, lifetime incidence of cardiomyopathy, heart failure, or major ventricular arrhythmia diagnosis was lower in participants with *RBM20* variants than in those with *TTN*tvs (hazard ratio, 0.55; 95% CI, 0.36-0.84; $P < .001$). Patients with *RBM20*tvs and DCM presented to referral centers later in life than those with P/LP *RBM20* and DCM (mean [SD], 53 [10] vs 34 [18] years; $P < .001$) and were less likely to have a family history of sudden cardiac arrest (2 of 10 [20%] vs 11 of 17 [65%]; $P = .046$) or cardiomyopathy (2 of 10 [20%] vs 14 of 18 [78%]; $P < .001$). There was no significant difference in age- and sex-adjusted incident major heart failure or arrhythmia events between patients with *RBM20*tv and DCM or those with P/LP *RBM20* and DCM, though sex-adjusted lifetime hazard was reduced in those with *RBM20*tv and DCM (hazard ratio, 0.13; 95% CI, 0.03-0.56; $P = .01$).

CONCLUSIONS AND RELEVANCE This study found that *RBM20* variants contributed to arrhythmogenic DCM phenotypes but conferred reduced lifetime disease penetrance compared to *TTN*tvs and milder disease severity alone than P/LP *RBM20* variants. Their potential for additive interactions with other damaging variants should be considered in patients with DCM and their families.

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Supplemental content

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Disease-causing variants in several dilated cardiomyopathy (DCM) genes have been associated with increased early risk of sudden cardiac arrest, prompting guideline recommendations for earlier consideration of defibrillator implant.[1,2] *RBM20* variants have been associated with such a highly penetrant arrhythmogenic phenotype, but only a small number of *RBM20* variants are known to cause DCM, all of which are missense variants that may have a dominant negative effect. Truncating variants in *RBM20* (*RBM20*tvs) have been identified in association with DCM in a small number of patients, but their comparative penetrance in the population and whether these variants carry a similar prognosis to disease-causing *RBM20* missense variants remains unknown. We investigated these questions in the genome-first UK Biobank (UKB) and All of Us populations, and an international cohort of patients with *RBM20* DCM.

Key Points

Question What is the contribution of *RBM20* truncating variants (*RBM20*tvs) to arrhythmogenic dilated cardiomyopathy (DCM)?

Findings In this cohort study including 4249 participants in UK Biobank and 179 in the RBM20 registry, *RBM20*tvs displayed overall low penetrance compared to a more common genetic cause of DCM (titin truncating variants) in large population biobanks. Patients with DCM and *RBM20*tvs had a milder lifetime disease course than those with known disease-causing variants in *RBM20*.

Meaning These observations suggest that *RBM20*tvs be viewed clinically as low-effect contributors to arrhythmogenic DCM.

Methods

Cohort Inclusion

RBM20 variants were identified from whole-genome sequences of participants in the UKB, All of Us, and from patients at 15 expert genetic DCM centers (the international *RBM20* DCM cohort). Variants in these cohorts, including pathogenic or likely pathogenic (P/LP), variants of uncertain significance (VUSs), and synonymous variants were centrally adjudicated by a licensed clinical genetic counselor and a board-certified medical geneticist (R.M. and B.F.) (eTable 6 in Supplement 2).

Participant consent for UKB inclusion is as documented previously.[4] Patients enrolled in the RBM20 registry were included after consent or waiver of consent as required by individual institutional review boards at each participating center. In the All of Us research program, informed consent for all participants is conducted in person or through an eConsent platform that includes primary consent, Health Insurance Portability and Accountability Act authorization for research electronic health records, and consent for return of genomic results. The protocol was reviewed by the institutional review board of the All of research program. The All of Us institutional review board follows the regulations and guidance of the National Institutes of Health Office for Human Research Protections for all studies. Analysis of the All of Us cohort was conducted with short-read whole-genome sequencing data from the All of Us controlled tier dataset version 8 (414 349 participants). Postsequencing variant and sample quality control were performed by the All of Us data and research center. Coding variants were extracted from Hail MatrixTables version 0.2.134 (Hail Team) and filtered for Phred quality score greater than 20, individual missingness less than 10%, minimum read coverage depth of 7, and presence in at least 1 participant passing the allele balance threshold of 0.20.

Variant Filtering and Diagnosis Definitions in UKB and All of Us

RBM20 and *TTN* variants with gnomAD v3 minor allele frequency less than 0.005 were identified among the UKB cohort of 502 132 participants with genome data using the UKB Browser.[3,4] For *RBM20*, all frameshift, nonsense, and synonymous variants were identified. The negative control population was defined as carriers of *RBM20* synonymous variants (excluding those with any SpliceAI score above 0). Titin truncating variants (*TTN*tvs) were identified as frameshift and nonsense variants within the A-band (GRCh38 2:178,535,982-178,618,491). Splice variants were excluded from both *RBM20* and *TTN* for this analysis given uncertainty regarding their heterogeneous effect. Genome variant files were obtained covering the exons and exon-intron junctions for *RBM20* and *TTN*. In total, 174 *RBM20* variant, 965 *TTN*tv, and 3110 *RBM20* synonymous variant carriers were identified (eTable 1 in Supplement 1).

Demographic and clinical data, including sex, birth year, birth month, age at enrollment, date of death, and dates of first reported diagnoses of cardiomyopathy (*International Statistical Classification of Diseases and Related Health Problems, Tenth Revision* [*ICD-10*] code I42), DCM (I42.0), cardiac arrest (I46), heart failure (HF; I50), atrial fibrillation and flutter (I48), other cardiac arrhythmias (I49), obesity (E66), hypertension (I10-I13 and I15), type 2 diabetes (E10-E14), and coronary artery disease (Z95.1, Z95.5, R93.1) were recorded. Three composite outcomes were evaluated: cardiomyopathy or HF diagnosis (codes I42, or I50), arrhythmia diagnosis (I46, or I49), and arrhythmogenic DCM diagnosis (I42, I46, I49, and I50). Atrial fibrillation and atrial flutter (I48) diagnoses were evaluated in a separate analysis.

Phenotypes in the All of Us cohort were selected to match those from the UKB set. Because All of Us includes *ICD-9* and *ICD-10* codes, the All of Us analysis included the *ICD-10* codes used for the UKB analysis plus the closest equivalent *ICD-9* codes.

Assembly, Variant Curation, and Event Definitions in International *RBM20* DCM Cohort

Patients with a variant in *RBM20* identified on clinical genetic testing were identified at international familial cardiomyopathy centers: Barts Heart Centre (London, UK), Brigham and Women's Hospital (Boston, Massachusetts, US), Children's Healthcare of Atlanta (Atlanta, Georgia, US), University of Colorado Anschutz Medical Campus (Aurora, Colorado, US), Heidelberg University (Heidelberg, Germany), Johns Hopkins University (Baltimore, Maryland, US), University of Pennsylvania

Table 1. Etiologic Fraction of *RBM20* Variants for Arrhythmogenic Dilated Cardiomyopathy (DCM) Diagnoses

Variant type	Cohort	OR (95% CI)	EF (95% CI)	*P* value[a]
Arrhythmogenic DCM diagnosis				
*RBM20*tv	UKB	2.15 (1.48 to 3.08)	0.53 (0.32 to 0.67)	<.001
*RBM20*sv	UKB	1.02 (0.91 to 1.13)	0.02 (−0.09 to 0.11)	.85
*TTN*tv	UKB	3.48 (3.06 to 3.98)	0.71 (0.67 to 0.75)	<.001
DCM diagnosis only				
*RBM20*tv	UKB	5.59 (1.78 to 17.51)	0.82 (0.44 to 0.94)	.003
*RBM20*sv	UKB	1.00 (0.91 to 1.13)	0.01 (−0.09 to 0.09)	.91
*TTN*tv	UKB	53.98 (42.12 to 69.18)	0.98 (0.95 to 0.98)	<.001

Abbreviations: EF, etiologic fraction; OR, odds ratio; sv, synonymous variant; tv, truncating variant; UKB, UK Biobank.

[a] Fisher exact test.

(Philadelphia, Pennsylvania, US), Royal Brompton Hospital (London, UK), Stanford University (Stanford, California, US), University of Trieste (Trieste, Italy), University of British Columbia (Vancouver, British Columbia, Canada), Victor Chang Cardiac Research Institute (Sydney, New South Wales, Australia), University of California, San Francisco (San Francisco, California, US), Wellspan Health (York, Pennsylvania, US), and University of Michigan (Ann Arbor, Michigan, US). Clinical data were collected retrospectively from real-world prospective care at these centers in case note abstraction forms. Dates were not provided but recorded as ages in years. Variables included baseline demographic characteristics, measurements from echocardiography, electrocardiogram, and cardiac magnetic resonance imaging, and clinical events occurring prior to initial evaluation and at the most recent follow up. Data were collected, stored, and shared in a deidentified manner in accordance with the institutional review board at each contributing center.

Variant pathogenicity was adjudicated centrally by a licensed clinical genetic counselor and a board-certified medical geneticist (R.M. and B.F.). Each variant was classified into 1 of 4 categories based on American College of Medical Genetics criteria with modifications for DCM: P/LP variants, VUSs, truncating variants, and benign/likely benign variants (eTable 6 in Supplement 2).[5] Individuals without genetic data, or who also carried a pathogenic variant in a different established cardiomyopathy gene listed by the ClinGen expert panel on dilated cardiomyopathies were excluded.[6] Of the 193 individuals enrolled with *RBM20* variants, 14 were excluded: 2 of these were due to the presence of a pathogenic variant in another cardiomyopathy gene, and the remaining 12 were excluded because they carried an *RBM20* VUS with ambiguous impact on splicing, which prevented further determination of whether the variant may lead to a truncated protein product. VUSs with ambiguous impact on splicing were identified based on the following characteristics: within 3 nucleotides of the exon-intron junction but with low SpliceAI score (<0.05; 5 individuals), in-frame insertions or deletions (4 individuals), intronic variants with borderline SpliceAI score (0.1-0.2; 2 individuals), and 1 individual with a multiexonic duplication involving the terminal exons.

Primary outcomes for survival analyses included 3 composite outcomes: (1) major HF events (defined by HF hospitalization (hospital admission with an HF-associated diagnosis), orthotopic heart transplantation, or left ventricular assist device implantation); (2) major ventricular arrhythmia events (defined by sustained ventricular tachycardia, appropriate implantable cardioverter defibrillator shock, sudden cardiac arrest or sudden cardiac death); and (3) combined major HF and major ventricular arrhythmia events. All deaths in this cohort occurred following a major ventricular arrhythmia event or HF event, and so death was not included in composite outcomes.

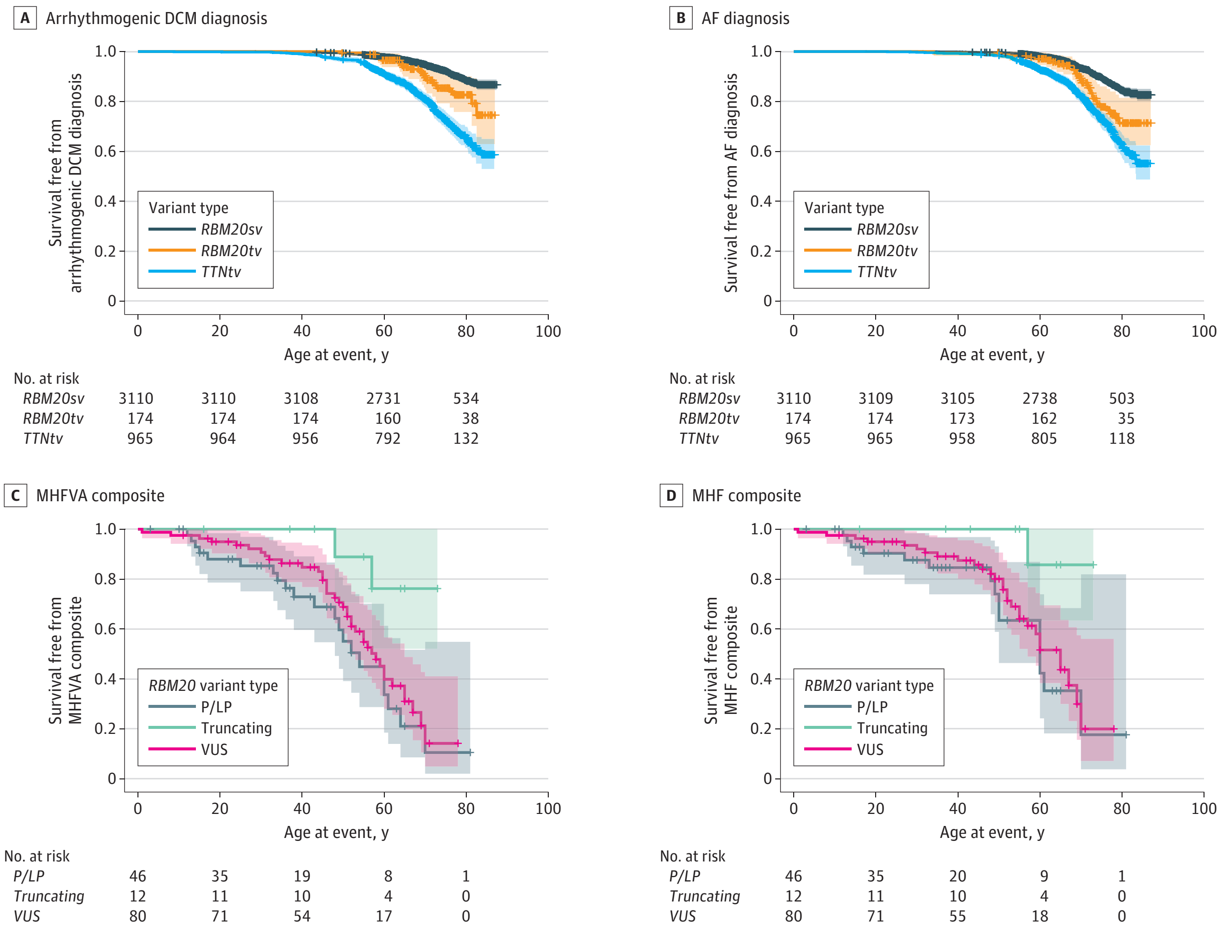
Statistical Analysis

Means were compared using *t* test and proportions were compared using χ^2 or Fisher exact test (for n < 100). Time-to-event data were compared using multivariate Cox regression analyses adjusting for age and/or sex. In all comparisons, patients with missing values for that comparison were excluded and curves were right trimmed at age 80 years. Etiologic fractions were calculated as a function of the odds ratio (OR: [OR − 1] / OR), to determine the attributable risk of *RBM20* variants to arrhythmogenic DCM (defined as above) or DCM phenotype (defined as *ICD-10* code I42.0 specifically).[7] For data collected at initial visit with a familial DCM center and baseline characteristics in UKB, means were compared using *t* test and proportions were compared using χ^2 or Fisher exact test (for n < 100). Time-to-event data from the International *RBM20* DCM cohort and in the UKB were compared at last follow-up using multivariate Cox regression analyses adjusting for age and/or sex as indicated in results. In all comparisons, patients with missing values for that comparison were excluded. Study dates are not disclosed because the institutional review board did not authorize the sharing of this information.

Results

Two main cohorts were studied for this project. In UK Biobank, a cohort of participants with *RBM20*tvs, *RBM20* synonymous variants, and *TTN*tvs was studied. Of these 4249 participants, 1869 (44%) were male. The mean (SD) age at enrollment was 56 (8.2) years. In the RBM20 registry, of 179 patients, 105 (58.6%) were male, and the mean (SD) age at enrollment was 43.8 (19.1) years. A validation cohort from the All of Us biobank was also used. This consisted of 7002 participants, 4342 of whom (62.0%) were male, and the mean (SD) age was 52.7 (16.7) years.

Figure. Kaplan-Meier Curves for *RBM20* Truncating Variant (*RBM20tv*) Penetrance in UK Biobank (UKB) and *RBM20tv* Survival Free From Major Heart Failure and Ventricular Arrhythmia in the *RBM20* Dilated Cardiomyopathy (DCM) Cohort



A, In UKB, lifetime incidence of arrhythmogenic DCM diagnoses compared to *RBM20* synonymous variants (hazard ratio [HR], 1.80; 95% CI, 1.22-2.79; *P* = .007) and titin truncating variant (*TTN*tv) carriers (HR, 0.51; 95% CI, 0.34-0.74; *P* < .001). B, In UKB, lifetime incidence of atrial fibrillation (AF) or flutter diagnoses compared to *RBM20* synonymous variants (*RBM20*sv; HR, 1.89; 95% CI, 1.32-2.71; *P* = .001) and *TTN*tv carriers (HR, 0.64; 95% CI, 0.42-0.91; *P* = .02). C, In the international *RBM20* DCM cohort, lifetime risk of combined major heart failure and ventricular arrhythmia (MHFVA) events in patients with *RBM20*tvs and *RBM20* variants of uncertain significance (VUSs) compared to patients with pathogenic or likely pathogenic (P/LP) *RBM20* variants (*RBM20*tv: HR, 0.13; 95% CI, 0.03-0.56; *P* = .01; *RBM20* VUS: HR, 0.79; 95% CI, 0.45-1.37; *P* = .40). D, In the international *RBM20* DCM cohort, lifetime risk of major heart failure events (MHF) alone in patients with *RBM20*tvs and *RBM20* VUSs compared to patients with P/LP *RBM20* variants (*RBM20*tv vs P/LP: HR, 0.10; 95% CI, 0.01-0.76; *P* = .03; *RBM20* VUS vs P/LP HR, 0.83; 95% CI, 0.44-1.58; *P* = .60). All analyses adjusted for sex.

Etiologic Fraction of *RBM20* Variants in Arrhythmogenic DCM

RBM20 variants were mildly enriched in participants with arrhythmogenic DCM-relevant diagnoses as compared to participants without arrhythmogenic DCM-relevant diagnoses in UKB (OR, 2.15; 95% CI, 1.48-3.08; etiologic fraction [EF], 0.53; 95% CI, 0.32-0.67; *P* < .001) (**Table 1**; eTable 1 in Supplement 1). *RBM20* variant enrichment in cardiomyopathy diagnoses specifically was higher (OR, 5.59; 95% CI, 1.78-17.51; EF, 0.82; 95% CI, 0.44-0.94; *P* = .003), though it remained much lower than the etiologic fraction of *TTN*tvs in UKB (*TTN*tv: OR, 53.98; 95% CI, 42.12-69.18; EF, 0.98; 95% CI, 0.95-0.98; *P* < .001). These results were similar in All of Us, with the exception *RBM20* variants associated with arrhythmogenic DCM diagnosis, which did not reach statistical significance (eTables 2-4 in Supplement 1). The transcript location of *RBM20*tv was not associated with disease ($\chi^2_1 = 0.05$; *P* = .82) (eFigure 1 in Supplement 1).

*RBM20*tv Penetrance in Large Biobanks

We assessed the lifetime risk of *RBM20*tvs in developing arrhythmogenic DCM-associated diagnoses. In UKB, no *RBM20* P/LP variants were present. We therefore compared participants with *RBM20* variants (n = 174) to those carrying *RBM20* synonymous variants (n = 3110) and those with P/LP *TTN*tvs (n = 965), the most common genetic cause of DCM (eTable 1 in Supplement 1). *RBM20*tv carriers experienced an increased lifetime hazard of the composite diagnosis compared to *RBM20* synonymous variant carriers (hazard ratio [HR], 1.80; 95% CI, 1.22-2.79; *P* = .007) but decreased compared with *TTN*tv carriers (HR, 0.51; 95% CI, 0.34-0.74; *P* < .001) (**Figure**, A). This finding was driven by HF and cardiomyopathy diag-

Table 2. Baseline Demographic Characteristics of the *RBM20* International Registry

Characteristic	P/LP missense *RBM20*	*RBM20*tv	*P* value
Individuals enrolled, No.	50	12	
Sex, No. (%)			
Male	26 (57)	8 (67)	.74
Female	24 (43)	4 (33)	
Probands, No. (%)	20 (40)	10 (83)	.01
Age at presentation, mean (SD), y			
All	31 (17)	48 (16)	.002
Probands	34 (18.X)	53 (9.9)	.004
FH SCA/D in probands, probands, No. (%)	11 (65)	2 (20)	.046
FH HF in probands, probands, No. (%)	14 (78)	2 (20)	.005
LVIDD, mean (SD), mm[a]	51	57	.22
LVIDD indexed, mean (SD), mm/m^2[a]	27	29	.58
LVEF, mean (SD), %[a]	42 (16)	41 (19)	.77
QRS, mean (SD), ms[b]	94 (13)	119 (29)	.005
QTc, mean (SD), ms[b]	416 (31.5)	430 (35.7)	.28
Prevalent LBBB, all, No. (%)[b]	0	3 (25)	<.001
Prevalent HF admission, all, No. (%)	9 (16)	1 (8)	.38
Prevalent ESHF, all, No. (%)	4 (8)	0	.32
Prevalent MHFE, all, No. (%)	9 (16)	1 (8)	.38
Prevalent MVA, all, No. (%)	6 (13)	1 (8)	.66
Prevalent MAHE, all, No. (%)	14 (31)	2 (17)	.36

Abbreviations: ESHF, end-stage heart failure; FH, family history; HF, heart failure; LA, left atrium; LBBB, left bundle branch block; LVEF, left ventricular ejection fraction; LVIDD, left ventricular internal diameter at end diastole; MAHE, major adverse arrhythmic and heart failure events; MHFE, major heart failure events; MVA, major ventricular arrhythmia; P/LP, pathogenic/likely pathogenic; SCA/D, sudden cardiac arrest/death; tv, truncating variant.

[a] Data from presenting echocardiogram.

[b] Data from presenting electrocardiograms.

noses (vs synonymous variants: HR, 2.14; 95% CI, 1.31-3.49; *P* = .002; vs *TTN*tvs: 0.46; 95% CI, 0.28-0.74; *P* < .001) but was not significantly different for arrhythmias (eFigure 2 in Supplement 1). Life-long incidence of atrial fibrillation or flutter was increased in *RBM20*tv carriers compared to synonymous variant carriers (HR, 1.89; 95% CI, 1.32-2.71; *P* = .001), and decreased compared to *TTN*tv carriers (HR, 0.64; 95% CI, 0.42-0.91; *P* = .02) (Figure, B). These sex-adjusted findings were similar in a smaller cohort of *RBM20*tv and *TTN*tv participants in All of Us[8], though were underpowered to detect statistical significance between *RBM20* variants and *TTN*tvs with respect to lifetime penetrance of arrhythmogenic DCM diagnoses, or between *RBM20* variants and synonymous variants with respect to atrial fibrillation or flutter and HF diagnosis (note that events occurring in populations <20 from UKB cannot be reported individually) (eFigure 3 and eTables 2-4 in Supplement 1).

Disease Course *of RBM20* Variants Compared to P/LP *RBM20* Missense Variants in DCM

Full baseline characteristics for this cohort appear in **Table 2.** *RBM20*tv probands presented later in life than *RBM20* P/LP probands, (mean [SD] age in those with *RBM20* variants: 53 [10] years, *RBM20* P/LP: 34 [18] years; *P* < .001) and *RBM20* VUS (*RBM20* VUS: 42 [18] years; *P* = .03), with a reduced frequency of family history of sudden cardiac arrest or DCM compared to *RBM20* P/LP probands (sudden cardiac arrest: 2 of 10 [20%] vs 11 of 17 [65%]; *P* = .046; DCM: 2 of 10 [20%] vs 14 of 18 078%]; *P* < .001) (Table 1; eTable 5 in Supplement 1). Patients with DCM and *RBM20*tvs, *RBM20* P/LP variants, and *RBM20* VUSs had a mean (SD) 5.0 (4.8), 7.3 (6.6), and 5.2 (5.4) years of follow-up after presentation to the contributing center, respectively. Age- and sex- adjusted comparison of incident composite HF and major ventricular arrhythmia events did not reveal a statistically significant difference between patients with DCM and *RBM20*tvs vs *RBM20* P/LP (*RBM20*tv: HR, 0.37; 95% CI, 0.04-3.18; *P* = .37; *RBM20* VUS: 0.33; 95% CI, 0.12-0.97; *P* = .04) (eFigure 4A-C in Supplement 1). Sex-adjusted comparison of lifetime composite HF and major ventricular arrhythmia events displayed a significant reduction of risk for patients with *RBM20*tvs but not those with *RBM20* VUSs compared to those with *RBM20* P/LP variants (*RBM20*tv: HR, 0.13; 95% CI, 0.03-0.56; *P* = .01; *RBM20* VUS: 0.79; 95% CI, 0.45-1.37; *P* = .39) (Figure, C). This difference was driven by a statistically significant reduction in lifetime risk of major HF events (*RBM20*tv: 0.10; 95% CI, 0.01,-0.76; *P* = .03; *RBM20* VUS: 0.83; 95% CI, 0.44-1.62; *P* = .60) (Figure, D). No statistically significant difference was identified in lifelong major ventricular arrhythmias (*RBM20*tv: 0.17; 95% CI, 0.02-1.34; *P* = .09; *RBM20* VUS: 0.53; 95% CI, 0.24-1.15; *P* = .11) (eFigure 4D in Supplement 1).

Discussion

This cohort study set out to evaluate potential disease contributions and relative severity of DCM associated with classes of variants like *RBM20* variants. Understanding such contributions will expand the utility of genetic testing for patients with DCM. For example, truncating variants, the most common cause of genetic DCM, have not been associated in large cohorts with as severe outcomes as DCM due to other genes (eg, *LMNA*).[9-12] Here we find that *RBM20* variants were associated with even less severe disease course than *TTN*tvs. Combined with the small etiologic fraction of arrhythmogenic DCM and DCM alone explained by *RBM20* variants, we would suggest that *RBM20* variants be viewed clinically as low-effect contributors to disease (ie, disease modifiers). This is helpful prognostic information for patients with mild DCM or incidentally identified *RBM20* variants and will help to prevent the erroneous assignation of poor prognosis to *RBM20*tv patients based on what is known about the severe course of patients with P/LP *RBM20* missense variants.[2,13,14]

Limitations

The findings of this study exist in the context of its necessary limitations. Studying patients referred to tertiary care centers introduces selection bias for a sicker population. We therefore also studied genome-first cohorts (UKB and All of Us), which select for less morbidity. The absolute estimates of etiologic fraction we make using diagnosis codes may underestimate absolute effect size. The sample size of patients with *RBM20*tv and DCM is small despite the large number of participant institutions, underlining the need for collaborative study in rare disease.

Conclusions

The findings in this cohort study represent a step in understanding the disease contribution of *RBM20* variants to arrhythmogenic DCM and interpreting their impact in the cardiovascular genetics clinic. Even for *RBM20* P/LP variants, risk factors for major ventricular arrhythmia events remain unknown beyond mild to moderate reductions in left ventricular ejection fraction.[2] Larger international efforts are needed to understand the risk structure of ventricular arrhythmias in *RBM20* cardiomyopathy, and whether *RBM20* variants confer additive risk to other genetic causes of DCM.

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